

Runbook: MCO User Lock / Restore — PRODUCTION

Item	Value
Script	Set-McoUserLock-PROD.ps1
Purpose	Temporarily disable all active MCO users (option 1) and restore them to their original values afterwards (option 2)
Database	Oracle 11g, host 172.20.3.16 , port 1526 , SID MCOPROD
Table	MC0001.UTILISATEUR
Excluded users (never changed)	PATKE01 , CICAM01 , KHARI01
Credential file	MCOPROD.cred (next to the script)
Backup file	UTILISATEUR_locked_users_PROD.csv (next to the script)
Runtime	Windows PowerShell 5.1
Environment	PRODUCTION — DEV equivalent: Set-McoUserLock.ps1 on MCODEV

Production change. Run only with an approved change ticket, in the approved window, after the same steps have been tested on MCODEV. Every save requires answering **Y** and typing MCOPROD .

1. What the script does

1.1 Option 1 — Disable active users

An **active user** is one where NBRE_TENTATIVE = 0 and COMPTE_VEROUILLE = 0 .

Step	Action
1	Checks that the backup file does not already exist. If it does, the script stops, so original values are never overwritten
2	Connects to 172.20.3.16:1526/MCOPROD and starts a transaction
3	Selects and locks (FOR UPDATE) every active user except the excluded codes
4	Lists the users and asks Disable N user(s) listed above? (default No)
5	On Yes : asks you to type MCOPROD (capitals). Anything else cancels with no change
6	Writes the original values to the backup CSV before saving any change

Step	Action
7	Sets <code>NBRE_TENTATIVE = 3</code> and <code>COMPTE_VEROUILLE = 1</code> for each listed user
8	Commits only if the number of rows updated matches the number listed; otherwise rolls back and deletes the backup

```
-- Selection
SELECT CODE_UTILISATEUR, NOM_UTILISATEUR, PRENOM_UTILISATEUR, NBRE_TENTATIVE, COMPTE_VEROUILLE
FROM MC0001.UTILISATEUR
WHERE NBRE_TENTATIVE = 0 AND COMPTE_VEROUILLE = 0
AND UPPER(TRIM(CODE_UTILISATEUR)) NOT IN ('PATKE01', 'CICAM01', 'KHARI01')
FOR UPDATE;

-- Update (per user)
UPDATE MC0001.UTILISATEUR
SET NBRE_TENTATIVE = 3, COMPTE_VEROUILLE = 1
WHERE CODE_UTILISATEUR = :code AND NBRE_TENTATIVE = 0 AND COMPTE_VEROUILLE = 0;
```

1.2 Option 2 — Restore users

Step	Action
1	Checks that the backup file exists
2	Lists the users in the backup and asks Restore N user(s) listed above to their original values? (default No), then asks you to type <code>MCOPROD</code>
3	For each user, sets <code>NBRE_TENTATIVE</code> and <code>COMPTE_VEROUILLE</code> back to their original values, only if they still hold <code>3</code> and <code>1</code>
4	Users changed by someone else since the lock are skipped and listed in a warning
5	Commits all restores in one transaction
6	Renames the backup to <code>UTILISATEUR_locked_users_PROD.restored_yyyyMMdd_HHmss.csv</code> so it can't be applied twice

```
UPDATE MC0001.UTILISATEUR
SET NBRE_TENTATIVE = :origN, COMPTE_VEROUILLE = :origC
WHERE CODE_UTILISATEUR = :code AND NBRE_TENTATIVE = 3 AND COMPTE_VEROUILLE = 1;
```

1.3 Backup file format

CSV with a header row, UTF-8. One row per user changed by option 1.

Column	Content
<code>CODE_UTILISATEUR</code>	User code
<code>NOM_UTILISATEUR</code> / <code>PRENOM_UTILISATEUR</code>	Last name / first name
<code>ORIG_NBRE_TENTATIVE</code> / <code>NEW_NBRE_TENTATIVE</code>	Value before (0) / after (3)

Column	Content
NBRE_TENTATIVE_CHG	Y if the column was changed
ORIG_COMPTE_VEROUILLE / NEW_COMPTE_VEROUILLE	Value before (0) / after (1)
COMPTE_VEROUILLE_CHG	Y if the column was changed
CHANGED_AT	Timestamp of the lock run

Do not edit, move or delete the backup file between option 1 and option 2. It is the only record of the original values.

2. Prerequisites

#	Requirement	How to check
1	Windows PowerShell 5.1	<code>\$PSVersionTable.PSVersion</code>
2	Network access to 172.20.3.16 on port 1526	<code>Test-NetConnection 172.20.3.16 -Port 1526</code> , which should return <code>TcpTestSucceeded : True</code>
3	Oracle account with <code>SELECT</code> and <code>UPDATE</code> on <code>MC0001.UTILISATEUR</code>	The same login can run an <code>UPDATE</code> in SQL Developer
4	Oracle.ManagedDataAccess.dll version 19.18.0 in the same folder as the script	See section 3.1

3. One-time setup

3.1 Install the Oracle driver (19.18.0)

Skip this step if the DLL is already in the folder (e.g. from the AGPH extract setup).

```
cd $env:USERPROFILE\Downloads
[Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
Invoke-WebRequest https://www.nuget.org/api/v2/package/Oracle.ManagedDataAccess/19.18.0 -OutFile odp.zip
Expand-Archive odp.zip -DestinationPath odp -Force
Copy-Item .\odp\lib\net40\Oracle.ManagedDataAccess.dll .
Unblock-File .\Oracle.ManagedDataAccess.dll
Unblock-File .\Set-McoUserLock-PROD.ps1
```

Use 19.18.0, not 23.x. Version 23.x needs extra .NET files that aren't on a standard Windows PC.

3.2 Save the Oracle credential

The script handles this itself. On the first run it asks for the Oracle user name and password and saves them to `MCOPROD.cred`. Later runs load the file without prompting.

Rule	Reason
Run the script as the Windows user who will use it from now on	The file is encrypted with Windows DPAPI: only that user, on that machine, can decrypt it
Run with <code>-ResetCred</code> whenever the Oracle password changes	Otherwise you get <code>ORA-01017</code>
Never copy the <code>.cred</code> file to another PC or user	It won't decrypt there

4. Run procedure

4.1 Pre-checks (before option 1)

Check	Action
Change approved	Approved change ticket for MCOPROD; you are inside the approved window
Tested in DEV	The same run (option 1 then option 2) completed successfully on MCODEV
Rollback ready	Option 2 is the rollback. Know where the backup CSV will be written (script folder)
No leftover backup	<code>Test-Path .\UTILISATEUR_locked_users_PROD.csv</code> must return <code>False</code>
Expected volume	In SQL Developer: <code>SELECT COUNT(*) FROM MC0001.UTILISATEUR WHERE NBRE_TENTATIVE = 0 AND COMPTE_VEROUILLE = 0 AND CODE_UTILISATEUR NOT IN ('PATKE01', 'CICAM01', 'KHARI01');</code> — note the number
Users informed	Users have been told they will be unable to log in

4.2 Option 1 — Disable active users

```
cd $env:USERPROFILE\Downloads
.\Set-McoUserLock-PROD.ps1 -Option 1
```

Expected output:

```
Using credentials for MC0001 from C:\Users\<you>\Downloads\MCOPROD.cred
Connected to PRODUCTION 172.20.3.16:1526/MCOPROD
```

CODE_UTILISATEUR	NOM_UTILISATEUR	PRENOM_UTILISATEUR	ORIG_NBRE_TENTATIVE	ORIG_COMPTE_VEROUILLE
-----	-----	-----	-----	-----
ABCDE01	...	...	0	0
...				

```
Confirm
Disable <N> user(s) listed above?
[Y] Yes [N] No [?] Help (default is "N"): Y
*** PRODUCTION: 172.20.3.16:1526/MCOPROD ***
Type MCOPROD to confirm: MCOPROD
Disabled <N> user(s). Originals saved to C:\Users\
<you>\Downloads\UTILISATEUR_locked_users_PROD.csv
```

<N> must match the pre-check count in 4.1. If it doesn't, answer **N** (or type anything other than MCOPROD) and investigate. Keep the backup CSV safe until option 2 has run — it is the only record of the original values.

4.3 Option 2 — Restore users

```
cd $env:USERPROFILE\Downloads
.\Set-McoUserLock-PROD.ps1 -Option 2
```

Expected output:

```
Using credentials for MC0001 from C:\Users\<you>\Downloads\MCOPROD.cred
Connected to PRODUCTION 172.20.3.16:1526/MCOPROD
...
Restore <N> user(s) listed above to their original values?
[Y] Yes [N] No [?] Help (default is "N"): Y
*** PRODUCTION: 172.20.3.16:1526/MCOPROD ***
Type MCOPROD to confirm: MCOPROD
Restored <N> user(s).
Backup archived to C:\Users\
<you>\Downloads\UTILISATEUR_locked_users_PROD.restored_20261006_153012.csv
```

If some users were changed after the lock, you also see:

```
WARNING: Skipped 2 user(s) - not found or values changed since lock: ABCDE01, FGHIJ02
```

Review those users manually (section 5).

4.4 Interactive menu

Running the script with no `-option` shows a menu:

*** PRODUCTION ***

MC0001.UTILISATEUR on 172.20.3.16:1526/MCOPROD

1) Disable all active users (except PATKE01, CICAM01, KHARI01)

2) Restore users from C:\Users\<you>\Downloads\UTILISATEUR_locked_users_PROD.csv

Select option (1/2):

4.5 Parameters

Parameter	Default	Use
-Option	(menu)	1 = disable, 2 = restore
-CredFile	<script folder>\MCOPROD.cred	Saved credential file
-ResetCred	Off	Prompt again and overwrite the .cred file
-UserName	(prompt)	Oracle user name used when creating the .cred file
-Credential	(from .cred)	Pass a credential directly; bypasses the .cred file
-BackupFile	<script folder>\UTILISATEUR_locked_users_PROD.csv	Where option 1 writes and option 2 reads the original values
-ExcludeUsers	PATKE01, CICAM01, KHARI01	User codes never changed by option 1
-DbHost	172.20.3.16	Database host
-Port	1526	Listener port
-Sid	MCOPROD	SID, or the service name when used with -UseServiceName
-UseServiceName	Off	Connect with SERVICE_NAME instead of SID
-Schema	MC0001	Schema owning UTILISATEUR
-DllPath	<script folder>\Oracle.ManagedDataAccess.dll	Driver location

5. Post-run verification

After	Check (SQL Developer)	Pass criteria
Option 1	SELECT COUNT(*) FROM MC0001.UTILISATEUR WHERE NBRE_TENTATIVE = 0 AND COMPTE_VEROUILLE = 0;	Only the excluded users remain (≤ 3)

After	Check (SQL Developer)	Pass criteria
Option 1	<code>SELECT CODE_UTILISATEUR, NBRE_TENTATIVE, COMPTE_VEROUILLE FROM MC0001.UTILISATEUR WHERE CODE_UTILISATEUR IN ('PATKE01', 'CICAM01', 'KHARI01');</code>	Unchanged
Option 1	Backup file	Exists; row count (excluding header) equals the disabled count
Option 2	Same <code>COUNT(*)</code> as option 1	Equals the pre-check count from 4.1 plus the excluded active users
Option 2	Skipped users (if any)	Each one reviewed and fixed manually or confirmed as intended
Option 2	Backup file	Renamed to <code>*.restored_<timestamp>.csv</code>

6. Exit codes

Code	Meaning
0	Finished: changes committed, cancelled at the prompt, or nothing to do
1	Failed. The error is printed (e.g. Failed: ORA-...). All changes in that run are rolled back

7. Troubleshooting

Symptom	Cause	Fix
Backup file '...' already exists (option 1)	A previous lock has not been restored	Run option 2 first, or use <code>-BackupFile</code> with a new name if the old file is no longer needed
Backup file '...' not found (option 2)	Wrong folder, or already restored	Run from the folder holding the CSV, or pass <code>-BackupFile</code> . Check for a <code>*.restored_*.csv</code>
Cancelled. No changes made. after typing the name	Typed text was not exactly <code>MCOPROD</code> (case-sensitive)	Run again and type <code>MCOPROD</code> in capitals
Script waits after Connected to ...	Another session holds uncommitted changes on <code>UTILISATEUR</code> rows	Ask the other session (or KYNDRYL Oracle Database Support) to commit or roll back; or press Ctrl+C and retry later
Skipped N user(s) ... (option 2)	Those users were changed after the lock (e.g. reset by support)	Check each user in SQL Developer and fix manually if needed
Expected N rows updated, got M. Rolled back.	Rows changed between select and update	Re-run option 1; nothing was saved

Symptom	Cause	Fix
Cannot read '...MCOPROD.cred' / Key not valid for use in specified state	.cred created by another Windows user or PC	Run with -ResetCred
ORA-01017: invalid username/password	Wrong or changed password	Run with -ResetCred
ORA-28000: account is locked	Oracle account locked	Ask KYNDRYL Oracle Database Support to unlock it
ORA-00942: table or view does not exist	No grant on MC0001.UTILISATEUR, or wrong schema	Ask KYNDRYL Oracle Database Support for SELECT / UPDATE ; check -Schema
ORA-01031: insufficient privileges	SELECT only, no UPDATE	Ask KYNDRYL Oracle Database Support for UPDATE on MC0001.UTILISATEUR
ORA-12505: listener does not currently know of SID	MCOPROD is a service name, not a SID	Add -UseServiceName
ORA-12170 / ORA-12541 / timeout	Network or firewall	Test-NetConnection 172.20.3.16 -Port 1526 ; check VPN; if blocked, submit a Firewall Request via ServiceNow
Oracle.ManagedDataAccess.dll not found	Driver missing	Section 3.1, or pass -DllPath
... running scripts is disabled on this system	Execution policy / downloaded file	Unblock-File .\Set-McoUserLock-PROD.ps1, or powershell.exe - ExecutionPolicy Bypass -File .\Set-McoUserLock-PROD.ps1

8. Security and housekeeping

Item	Guidance
Credentials	Only in the DPAPI-encrypted MCOPROD.cred ; never in the script, command history or email
Backup CSV	Production data: contains user codes and names. Attach a copy to the change ticket as evidence. Keep it on a bank-managed drive; never commit it to Git (it is in .gitignore)
Restored backups	Keep *.restored_*.csv as the audit trail per retention policy, then delete
Excluded users	Change only with approval; pass -ExcludeUsers rather than editing the script
Script changes	Keep under version control (kso1013/fscprd); test on MCODEV (Set-McoUserLock.ps1) before running in PROD

9. Contacts and escalation

Issue	Contact
Database access, grants, row locks	KYNDRYL — Oracle Database Support
Network or firewall to 172.20.3.16:1526	Please submit a Firewall Request via ServiceNow
MCO application / user impact	Kevin Paterson
Script defects	Kevin Paterson

10. Change log

Date	Change
2026-10-06	Initial PROD version (Set-McoUserLock-PR0D.ps1), from the tested MCODEV script: MCOPROD defaults, separate .cred and backup files, typed MCOPROD confirmation, red PRODUCTION banner